



| Connecting Strength

K2 Base Report

BDP-4863



Project address	71 Renningen
Company	TestProjectSharingGmbH
Address	Industrie str 18, 71272, Renningen
Website	www.dummy.com
Planner	shobana Test account
Issue date	02/06/2026
Version	K2 Base Version 3.2.94.0



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About us

K2 Systems. Innovative mounting system from a strong team.

Since 2004 we have been developing pioneering and highly functional mounting system solutions for photovoltaic installations around the world. Our systems are designed in our own product development department where we continually optimize and adapt mounting systems to the ever-changing market.

A knowledgeable and friendly team

Just like a mountain climbing team, K2 Systems is built on mutual trust. This applies to our customer service as well as within the company itself, because we believe a trusting partnership leads to successful photovoltaic projects.

Our employees place total focus on the needs and wishes of our customer. This is true in all company departments.

10 locations and worldwide sales network

In our international team, everyone works together to provide customers with competent, comprehensive and entirely personalized service.

This is especially true in the constant training our employees undergo with regards to product optimization, quality assurance, or innovations in construction techniques.

Quality management and certificates

K2 Systems stands for Connecting Strength, the highest quality, and precision-crafted and customized components. Our customers and business partners deeply appreciate all of these factors. Three independent authorities have tested, confirmed, and certified our skills and components. External authorities are not the only ones to have put K2 Systems to the test. Our internal quality control ensures that all our products are subject to a constant review process.

These measures all ensure the outstanding quality standards that exemplify products from K2 Systems, and which we maintain through largely exclusive "Made in Germany" or "Made in Europe" practices.



Product guarantee

K2 Systems offers a 12-year product warranty on all products in its integrated range. The use of high quality materials and a three-level quality inspection ensure these standards.

In a nutshell

As roof-top specialists, we offer effective and economical solutions for roofs all around the world and provide professional, fast and reliable support for our customers in the solar industry.

The static report is not including module and building verification.



Project overview

Roofs

Roof	System	Module	Height	Quantity	Total power
<u>Bereich 1</u>  Seam metal	<u>SolidRail</u>	AHT60M-250WBL 1,650×991×40 mm 250 Wp	15.00 m	132	33 kWp
<u>Bereich 2</u>  Bitumen / Composite Shingles	<u>SingleRail</u>	AHT60M-250WBL 1,650×991×40 mm 250 Wp	15.00 m	32	8 kWp
Total				164	41.00 kWp

Project information

Address	71 Renningen
Author	shobana Test account

Load settings

Design method	DIN EN
Failure consequence class (CC)	CC2
Design working life	50 years
Terrain category	I/II - mixed profile rural area
Wind load zone	1
Snow load zone	2
Snow load on ground level	1.23 kN/m ²

Material values

For material information refer to the product catalogue:

[K2 Catalogue \(k2-systems.com\)](https://k2-systems.com)



THE PROJECT IS VERIFIED.

The selected mounting system can be installed as planned
Thank you for choosing a K2 mounting system.

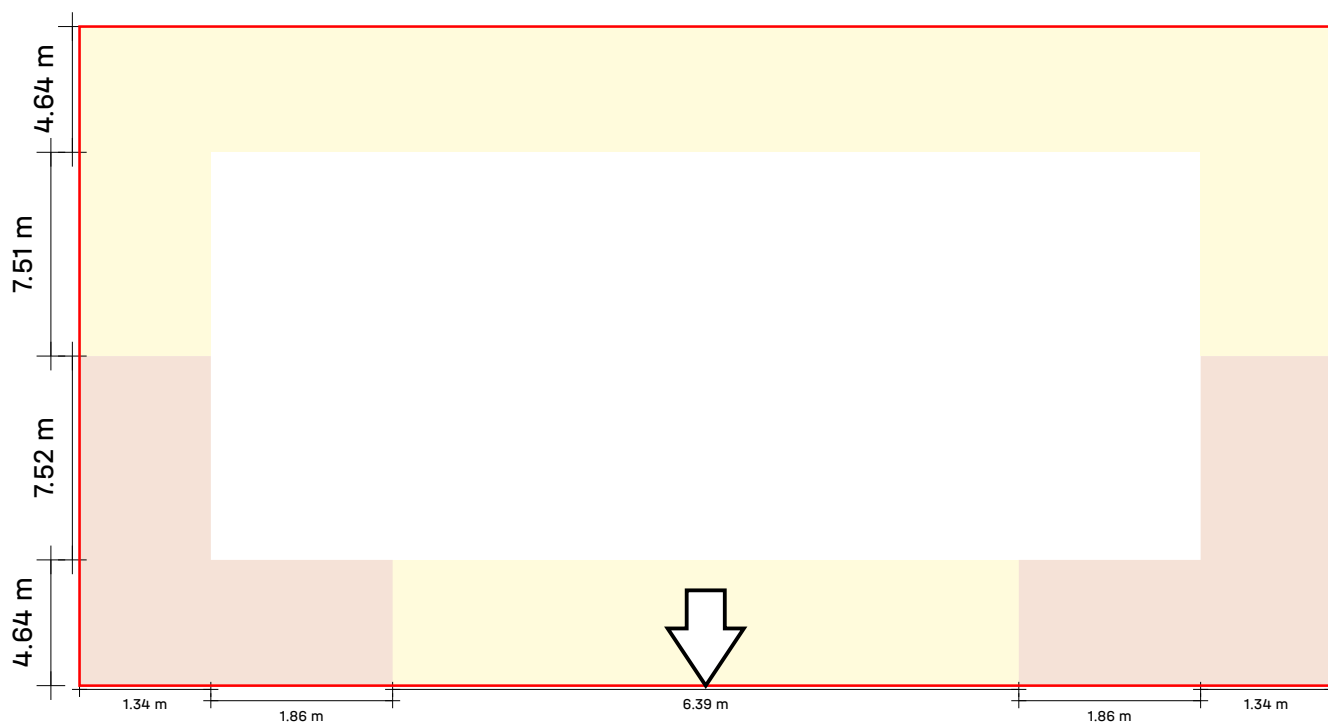
BDP-4863



Project information

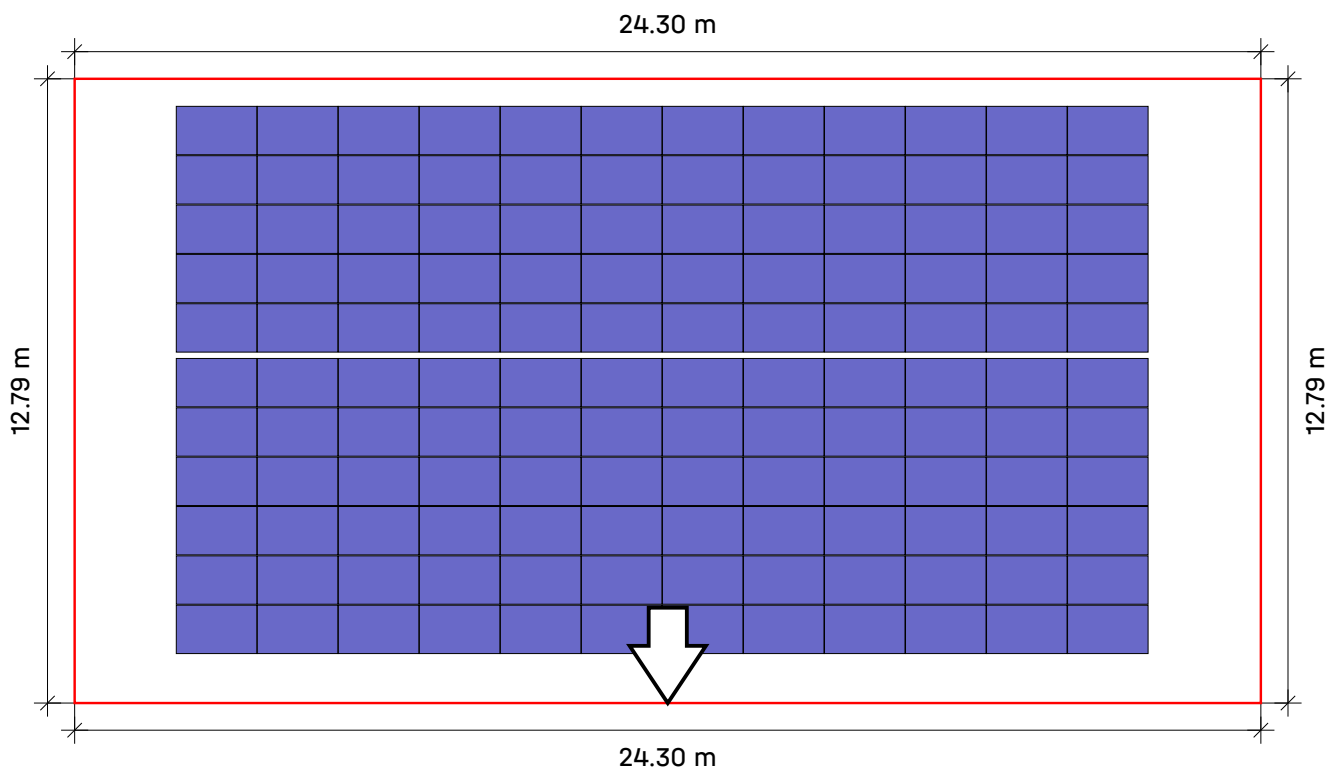
Address	71 Renningen
Author	shobana Test account

Bereich 1





Bereich 1



Roof	System	Module	Height	Quantity	Total power
<u>Bereich 1</u>  Seam metal	<u>SolidRail</u>	AHT60M-250WBL 1,650×991×40 mm 250 Wp	15.00 m	132	33 kWp



Bereich 1 | Assembly plan

Base Rails

Type	Whole Rails		Rail cutting		
	Total Rail Length	Quantity 4.40 m	Part of Rail / Rest	Length	Rest
7*A	19.360 m	4*4.40 m	4.400	1.760 from 4.400	<u>2.630</u>
6*B	19.360 m	4*4.40 m	<u>2.630</u>	1.760 from 2.630	0.860

1 cm is viewed as lost for each cutting

Red numbers are leftover rails which will not be used any longer

Upper Rail

Type	Whole Rails		Rail cutting		
	Total Rail Length	Quantity 2.10 m	Part of Rail / Rest	Length	Rest
24* A	6.146 m	2*2.10 m	2.100	1.946 from 2.100	0.144
12*B	5.135 m	1*2.10 m	2.100	2.035 from 2.100	0.055
			2.100	1.000 from 2.100	<u>1.090</u>
12*C	5.135 m	1*2.10 m	2.100	2.035 from 2.100	0.055
			<u>1.090</u>	1.000 from 1.090	0.080

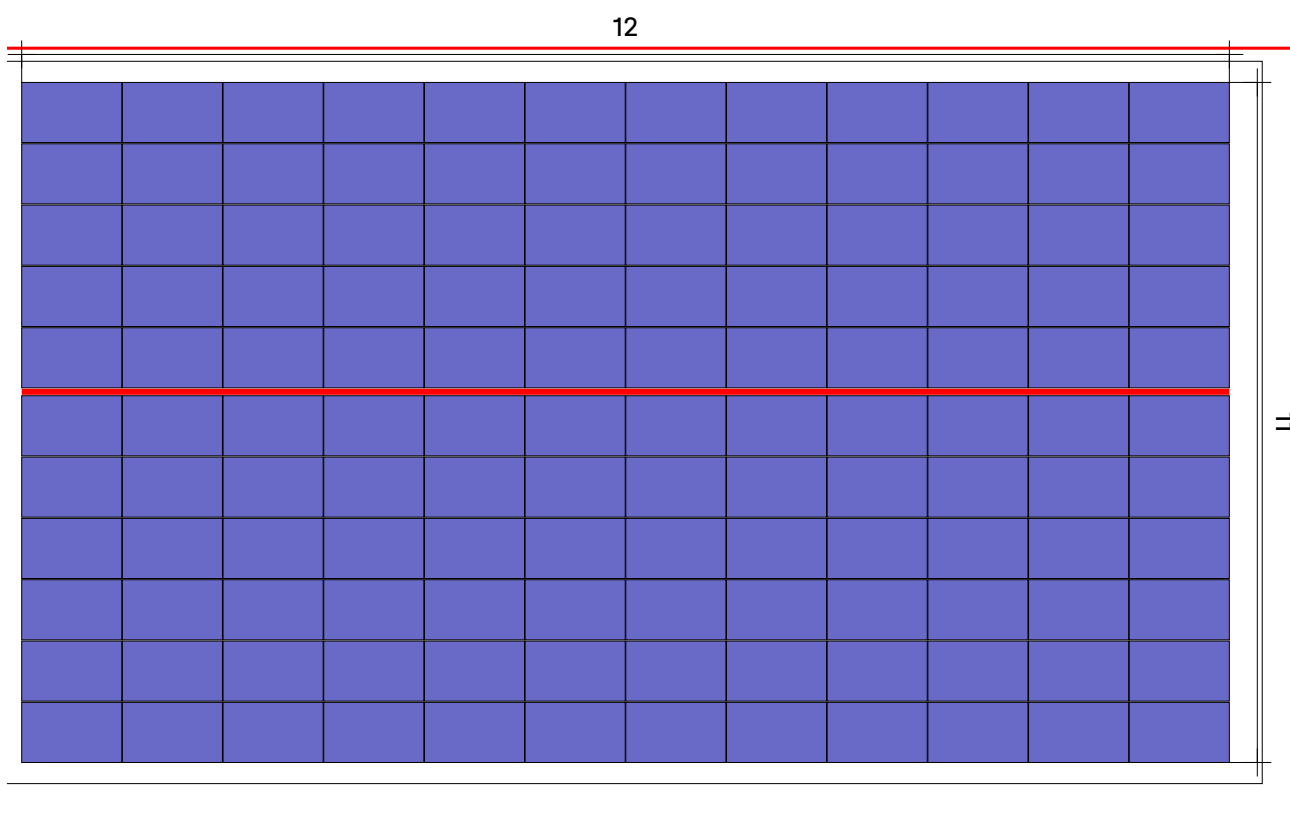
Fastener Spacing

Module	Array	Distance	maximum cantilever length [m]	maximum fastener spacing [m]
1	field area	0.41 m	0.406	0.000
1	ridge	0.41 m	0.406	0.000
1	eaves	0.41 m	0.406	0.000
1	gableboard	0.41 m	0.406	0.000
1	corner region (eave)	0.41 m	0.406	0.000

Module arrays

Module array	Width[m]	Length[m]	Width in modules	Length in modules
1	19.91	11.21	12	11

Bereich 1 | Module array 1



Roof ① Module array ①

Mounting System

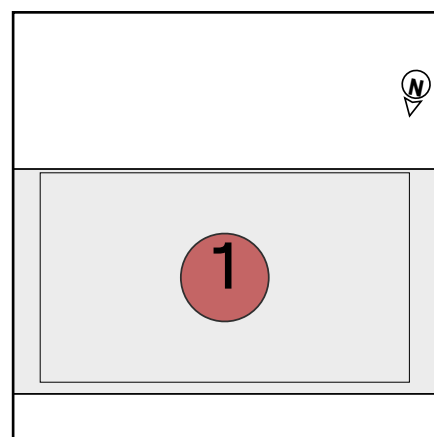
SolidRail

Module

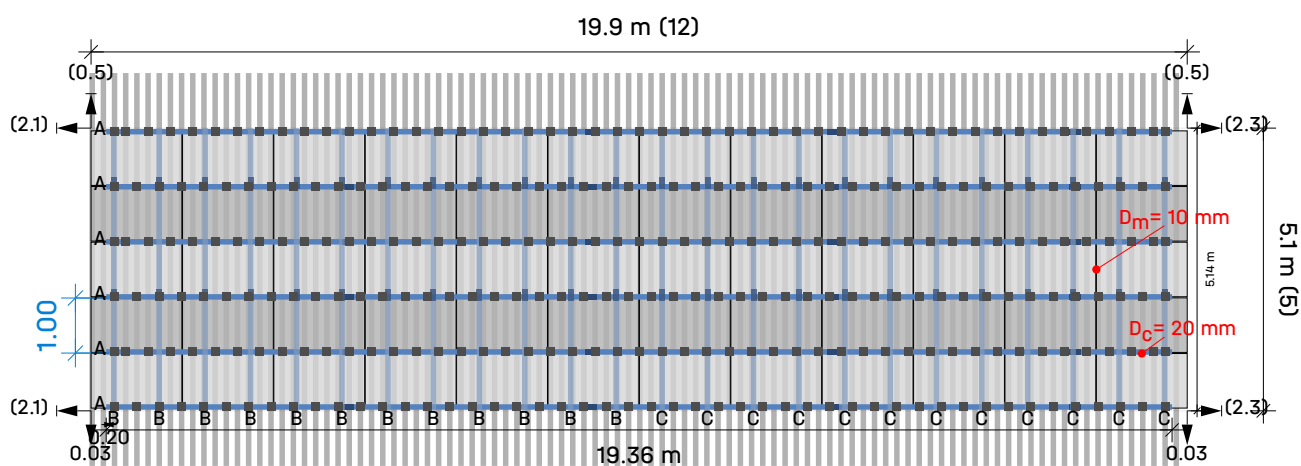
132(33 kWp) x
AHT60M-250WBL

Row spacing

1.66 m



Bereich 1 | Module array 1 | Module block 1

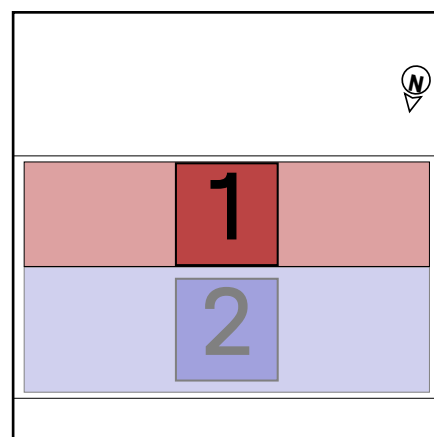


Roof ① Module array ① Module block 1

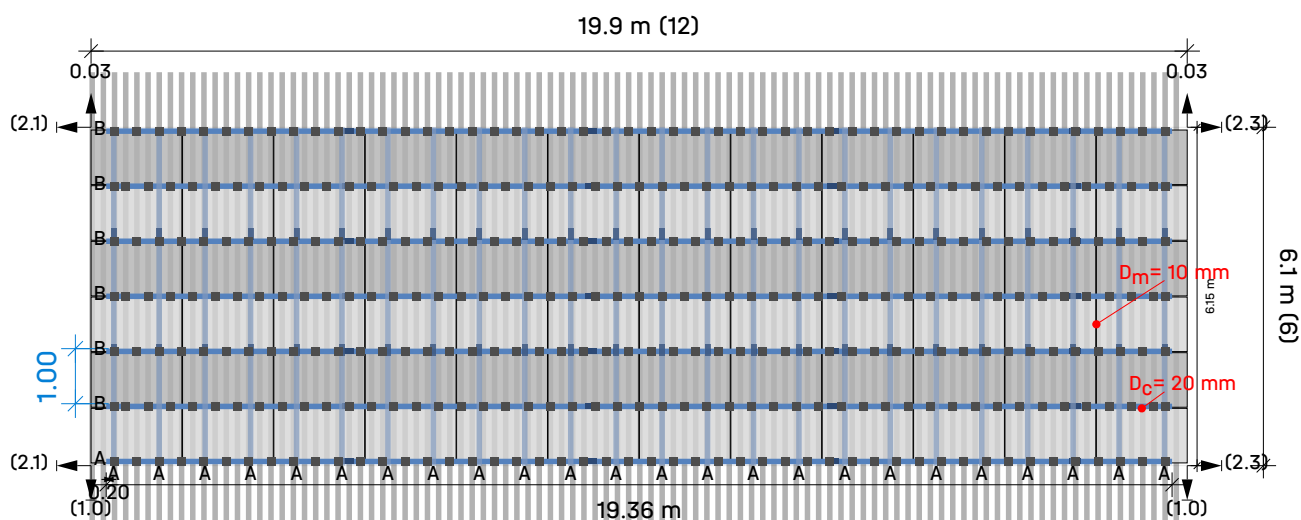
Modules $12 \times 5 = 60$

Legend

- Fastener
- Mounting rail: K2 SolidRail UltraLight 32
- Mounting rail (Upper): K2 SingleRail 36
- Rail Distance [m]
- Dist. to neighbour module block/array [m]
- D_c Distance for clamping between modules
- D_m Distance between modules



Bereich 1 | Module array 1 | Module block 2

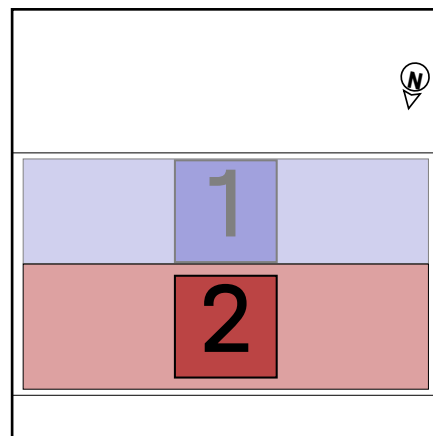


Roof ① Module array ① Module block 2

Modules $12 \times 6 = 72$

Legend

- Fastener
- Mounting rail: K2 SolidRail UltraLight 32
- Mounting rail (Upper): K2 SingleRail 36
- Rail Distance [m]
- Dist. to neighbour module block/array [m]
- D_c Distance for clamping between modules
- D_m Distance between modules





Results | Bereich 1

Roof	System	Module	Height	Quantity	Total power
<u>Bereich 1</u>  Seam metal	<u>SolidRail</u>	AHT60M-250WBL 1,650×991×40 mm 250 Wp	15.00 m	132	33 kWp

Module

Name	AHT60M-250WBL
Manufacturer	Ahead Renewable Energy Ltd.
Output power	250 Wp
Dimensions	1,650×991×40 mm
Weight	20.0 kg

Components

Fastener	Solid Clamp Rib-Roof Evolution
Base rails	K2 SolidRail UltraLight 32
Upper Rail	K2 SingleRail 36

Loads on modules (module dimensioning)

No. Module Array	Array	A-TrA [m²]	ultimate state [Pa]				Serviceability [Pa]			
			Pressure ⊥	Pressure 	Uplift ⊥	Uplift 	Pressure ⊥	Pressure 	Uplift ⊥	Uplift
1	field area	1.64	1,810.2	145.4	-1,203.8	12.5	1,221.1	98.2	-754.9	12.5
1	ridge	1.64	1,810.2	145.4	-1,203.8	12.5	1,221.1	98.2	-754.9	12.5
1	eaves	1.64	1,810.2	145.4	-2,110.2	12.5	1,221.1	98.2	-1,359.2	12.5
1	gableboard	1.64	1,810.2	145.4	-2,136.5	12.5	1,221.1	98.2	-1,376.8	12.5
1	corner region (eave)	1.64	1,810.2	145.4	-2,726.1	12.5	1,221.1	98.2	-1,769.8	12.5

Base Rails - Utilisation result

No. Module Array	roof areas	ultimate limit state			Usab.	Distances		maximum values	
		Pr σ[%]	CL σ[%]	Fst F[%]	Pr f[%]	Fst [m]	BR [m]	CL L _{max} [m]	Fst D _{max} [m]
1	field area	12.0	0.0	0.0	3.4	0.406	1.000	0.406	0.000
1	ridge	12.0	5.8	0.0	3.4	0.406	1.000	0.406	0.000
1	eaves	12.0	0.0	0.0	3.4	0.406	1.000	0.406	0.000
1	gableboard	12.0	8.4	0.0	3.4	0.406	1.000	0.406	0.000
1	corner region (eave)	12.4	8.7	0.0	3.5	0.406	1.000	0.406	0.000



Results | Bereich 1

Upper Rail - Utilisation result

No. Module Array	roof areas	ultimate limit state			Usab.	Distances		maximum values	
		Pr σ [%]	CL σ [%]	Fst F[%]	Pr f[%]	Fst [m]	UR [m]	CL L_{max} [m]	Fst D _{max} [m]
1	field area	38.6	1.0	---	24.3	1.000	---	0.481	1.576
1	ridge	38.6	0.8	---	24.3	1.000	---	0.481	1.576
1	eaves	38.6	1.1	---	24.3	1.000	---	0.481	1.576
1	gableboard	38.6	1.0	---	24.3	1.000	---	0.481	1.576
1	corner region (eave)	41.4	1.2	---	25.0	1.000	---	0.478	1.554

Pr	Profile	Fst D _{max}	maximum fastener spacing [m]
Fst	Fastener	BR	Base Rail
σ	Stress	UR	Upper rail
f	Deflection	Usab.	serviceability limit state
F	Force	CL	Cantilever
CL/ L_{max}	maximum cantilever length [m]		



Results | Bereich 1

Important Information

- Seam clamps are not to be positioned at the fastening clips of the roof sheeting.
- The assembly instruction of the seam clamp used needs to be regarded.
- When planning in Base, only the rails are verified for load-bearing capacity.
No static proof of the seam clamps on the existing roof is done.
Effects on the building can be determined by a civil engineer.
- The structure was statically verified in accordance with Eurocode 9: Design of aluminium structures (DIN EN 1999-1-1:2021) and offers sufficient load-bearing capacity and stability for the loads specified in the chapter 'Maximum actions on the components'.
- Adjustment factor for wind load regarding service life period, f_W , is according to DIN EN 1991-1-4/NA, NDP for 4.2 (2P) note 5, table 3
- Adjustment factor for snow load regarding service life period, f_S , is according to DIN EN 1991-1-3/annex D, table 4
- The design rules comply with the Eurocode EN 1990 - Basis of structural design. (DIN EN 1990: 2010)
- The determination of snow loads is based on the National Annex to DIN EN 1991-1-3/NA - Snow Loads.
- The determination of wind loads is based on the National Annex to DIN EN 1991-1-4/NA - Wind Loads.
- Service life is recognised according to 'Eurocode EN 1991 - Action on structures, Snow loads' and 'Eurocode EN 1991 - Actions on structures, Wind actions'. Subject to the Building Regulations and for security-relevant reasons the installation has to be dismantled at the end of its service life.
- Failure consequence class is considered according to 'Eurocode EN 1990 - Basis of structural design'.
- The person responsible for the execution of the work must check the load assumptions made with the conditions on site. If deviations are found, the person who prepared the static calculation must be consulted immediately.
- Please note our General Terms and Conditions of Use (TCU-E) in their currently valid version, available at: <https://k2-systems.com/en/digital-services/general-terms-and-conditions-of-use-for-entrepreneurs-tcu-e/>
Please note in particular § 1, Special provisions for K2 Base, clause 3 ("Technical and specialist requirements on the Customer's premises."), § 6 ("Limitation of warranty") and § 7 ("Limitation of liability").
- All instructions and requirements of the static calculations must be observed.
- Compliance with the regulations and requirements of the building permit and the building trade associations must be ensured during the execution of all work.
- All connections and anchoring measures must be carried out using approved and standardized connecting elements or parts only.
- Subordinate components that have not been verified must be implemented structurally.
- The "General Rules" for corrosion protection design of the structure are strongly recommended for application during implementation.
- The structure may only be climbed by trained and authorized personnel wearing appropriate safety equipment. Unauthorized use of the structure poses a risk of falling.



Structural analysis report | Bereich 1

General information

Name	BDP-4863
Mounting System	SolidRail
Author	shobana Test account

Location information

Address	71 Renningen
Ground elevation	405.51 m

Roof information

Building height	15.00 m
Roof type	Gable roof
Roof pitch	5°
Roof covering	Seam metal
Min. roof edge distance	0.00 m
Seam Type	Other
Seam distance	203.0 mm
Roof material	no selection
Sheet thickness	0.480 mm

Loads

Design method	DIN EN
Failure consequence class (CC)	CC2
Design working life	50 years
Terrain category	I/II - mixed profile rural area

Wind load

Wind load zone	1
Velocity pressure, 50 years	$q_{p,50} = 0.821 \text{ kN/m}^2$
Adjustment factor for service life	$f_w = 1.000$
Velocity pressure, 25 years	$q_{p,25} = 0.821 \text{ kN/m}^2$



Structural analysis report | Bereich 1

Roof areas

No. Module Array	Array	load impact area [m ²]	maxCpe	minCpe	wind pressure [kN/m ²]	wind suction [kN/m ²]
1	field area	10.00	0.200	-0.700	0.164	-0.575
1	ridge	10.00	0.200	-0.700	0.164	-0.575
1	eaves	10.00	0.200	-1.200	0.164	-0.985
1	gableboard	10.00	0.200	-1.300	0.164	-1.067
1	corner region (eave)	10.00	0.200	-1.700	0.164	-1.396

Snow load

Snow load zone	2
Snow guard	No
Snow load on ground level	$s_k = 1.234 \text{ kN/m}^2$
Shape Coefficient for Snow	$\mu_i = 0.800$
Factor for roof pitch	$d_i = 0.996$
Snow load on roof, 50 year	$s_{i,50} = 0.983 \text{ kN/m}^2$
Adjustment factor for service life	$f_s = 1.000$
Snow load on roof, 25 years	$s_{i,25} = 0.983 \text{ kN/m}^2$

Dead Load

Weight of module	$G_M = 20.0 \text{ kg}$
Weight of mounting system per module	$= 3.90 \text{ kg}$
Module area	$A_M = 1.64 \text{ m}^2$
Dead weight of module per m ²	$= 12.23 \text{ kg/m}^2$
Dead weight of mounting system per m ²	$= 2.39 \text{ kg/m}^2$
Total Dead Load (excl. ballast) per m ²	$= 0.14 \text{ kN/m}^2$



Structural analysis report | Bereich 1

Load Combinations

Ultimate limit state

Partial safety factor unfavourable permanent load	$\gamma_{G,sup} = 1.35$
Partial safety factor favourable permanent load	$\gamma_{G,inf} = 1.00$
Partial safety factor destabilising permanent load	$\gamma_{G,dst} = 1.10$
Partial safety factor stabilising permanent load	$\gamma_{G,stab} = 0.90$
Partial safety factor variable loads	$\gamma_Q = 1.50$
Combination coefficient with regards to wind	$\psi_{0,W} = 0.60$
Combination coefficient with regards to Snow	$\psi_{0,S} = 0.50$
Importance factor permanent	$\kappa_{Fl,G} = 1.00$
Importance factor variable	$\kappa_{Fl,Q} = 1.00$

Load case combination 01	$LCC\ 01_{uls} = \gamma_{G,sup} * \kappa_{Fl,G} * G_k + \gamma_Q * \kappa_{Fl,Q} * S_{i,n}$
Load case combination 02	$LCC\ 02_{uls} = \gamma_{G,sup} * \kappa_{Fl,G} * G_k + \gamma_Q * \kappa_{Fl,Q} * W_{k,Pressure}$
Load case combination 03	$LCC\ 03_{uls} = \gamma_{G,sup} * \kappa_{Fl,G} * G_k + \gamma_Q * \kappa_{Fl,Q} * (W_{k,Pressure} + \psi_{0,S} * S_{i,n})$
Load case combination 04	$LCC\ 04_{uls} = \gamma_{G,sup} * \kappa_{Fl,G} * G_k + \gamma_Q * \kappa_{Fl,Q} * (S_{i,n} + \psi_{0,W} * W_{k,Pressure})$
Load case combination 06	$LCC\ 06_{uls} = \gamma_{G,inf} * G_k + \gamma_Q * \kappa_{Fl,Q} * W_{k,Suction}$

Serviceability limit state

Combination coefficient with regards to wind	$\psi_{0,W} = 0.60$
Combination coefficient with regards to Snow	$\psi_{0,S} = 0.50$

Load case combination 01	$LCC\ 01_{sls} = G_k + S_{i,n}$
Load case combination 02	$LCC\ 02_{sls} = G_k + W_{k,Pressure}$
Load case combination 03	$LCC\ 03_{sls} = G_k + W_{k,Pressure} + \psi_{0,S} * S_{i,n}$
Load case combination 04	$LCC\ 04_{sls} = G_k + S_{i,n} + \psi_{0,W} * W_{k,Pressure}$
Load case combination 06	$LCC\ 06_{sls} = G_k + W_{k,Suction}$

Maximum load on modules (Mounting system dimensioning)

No. Module Array	Array	A-TrA [m ²]	ultimate state [kN/m ²]				Serviceability [kN/m ²]			
			Pressure ⊥	Pressure 	Uplift ⊥	Uplift 	Pressure ⊥	Pressure 	Uplift ⊥	Uplift
1	field area	10.00	1.810	0.145	-0.719	0.012	1.221	0.098	-0.432	0.012
1	ridge	10.00	1.810	0.145	-0.719	0.012	1.221	0.098	-0.432	0.012
1	eaves	10.00	1.810	0.145	-1.335	0.012	1.221	0.098	-0.843	0.012
1	gableboard	10.00	1.810	0.145	-1.458	0.012	1.221	0.098	-0.925	0.012
1	corner region (eave)	10.00	1.810	0.145	-1.951	0.012	1.221	0.098	-1.253	0.012

Structural analysis report | Bereich 1

Max. load on fastener

No. Module Array	Array	A-TrA [m ²]	ultimate state [kN]				Serviceability [kN]			
			Pressure ⊥	Pressure 	Uplift ⊥	Uplift	Pressure ⊥	Pressure 	Uplift ⊥	Uplift
1	field area	10.00	0.772	0.062	-0.307	0.005	0.521	0.042	-0.184	0.005
1	ridge	10.00	0.772	0.062	-0.307	0.005	0.521	0.042	-0.184	0.005
1	eaves	10.00	0.772	0.062	-0.569	0.005	0.521	0.042	-0.359	0.005
1	gableboard	10.00	0.772	0.062	-0.622	0.005	0.521	0.042	-0.394	0.005
1	corner region (eave)	10.00	0.772	0.062	-0.832	0.005	0.521	0.042	-0.534	0.005

Resistance Values of Components

Base Rails

Base Rails	A [cm ²]	I _y [cm ⁴]	I _z [cm ⁴]	W _y [cm ³]	W _z [cm ³]
K2 SolidRail UltraLight 32	2.610	2.59	5.54	1.57	2.84

Upper rail

Upper rail	A [cm ²]	I _y [cm ⁴]	I _z [cm ⁴]	W _y [cm ³]	W _z [cm ³]
K2 SingleRail 36	2.850	4.02	6.37	2.14	3.09

Base Rails - Utilisation result

No. Module Array	roof areas	ultimate limit state			Usab. Pr f[%]	Distances		maximum values	
		Pr σ[%]	CL σ[%]	Fst F[%]		Fst [m]	BR [m]	CL L _{max} [m]	Fst D _{max} [m]
1	field area	12.0	0.0	0.0	3.4	0.406	1.000	0.406	0.000
1	ridge	12.0	5.8	0.0	3.4	0.406	1.000	0.406	0.000
1	eaves	12.0	0.0	0.0	3.4	0.406	1.000	0.406	0.000
1	gableboard	12.0	8.4	0.0	3.4	0.406	1.000	0.406	0.000
1	corner region (eave)	12.4	8.7	0.0	3.5	0.406	1.000	0.406	0.000



Structural analysis report | Bereich 1

Upper Rail - Utilisation result

No.	roof areas	ultimate limit state			Usab.	Distances		maximum values	
		Pr	CL	Fst		Fst	UR	CL	Fst
Module Array		σ [%]	σ [%]	F[%]	f[%]	[m]	[m]	L_{max} [m]	Fst D_{max} [m]
1	field area	38.6	1.0	---	24.3	1.000	---	0.481	1.576
1	ridge	38.6	0.8	---	24.3	1.000	---	0.481	1.576
1	eaves	38.6	1.1	---	24.3	1.000	---	0.481	1.576
1	gableboard	38.6	1.0	---	24.3	1.000	---	0.481	1.576
1	corner region (eave)	41.4	1.2	---	25.0	1.000	---	0.478	1.554

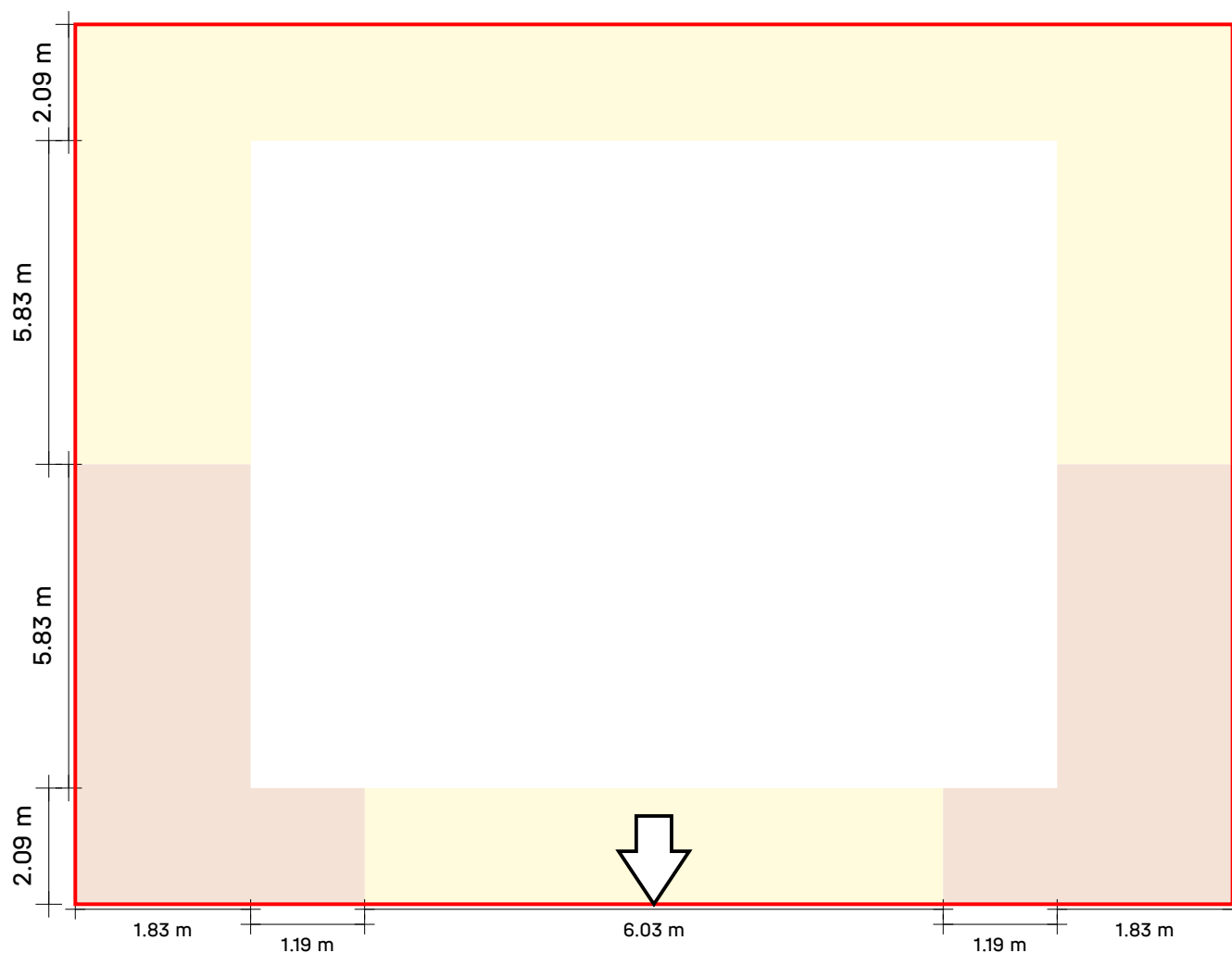
Pr	Profile	Fst D_{max}	maximum fastener spacing [m]
Fst	Fastener	BR	Base Rail
σ	Stress	UR	Upper rail
f	Deflection	Usab.	serviceability limit state
F	Force	CL	Cantilever
CL/L_{max}	maximum cantilever length [m]		



Bereich 1 | Bill of material

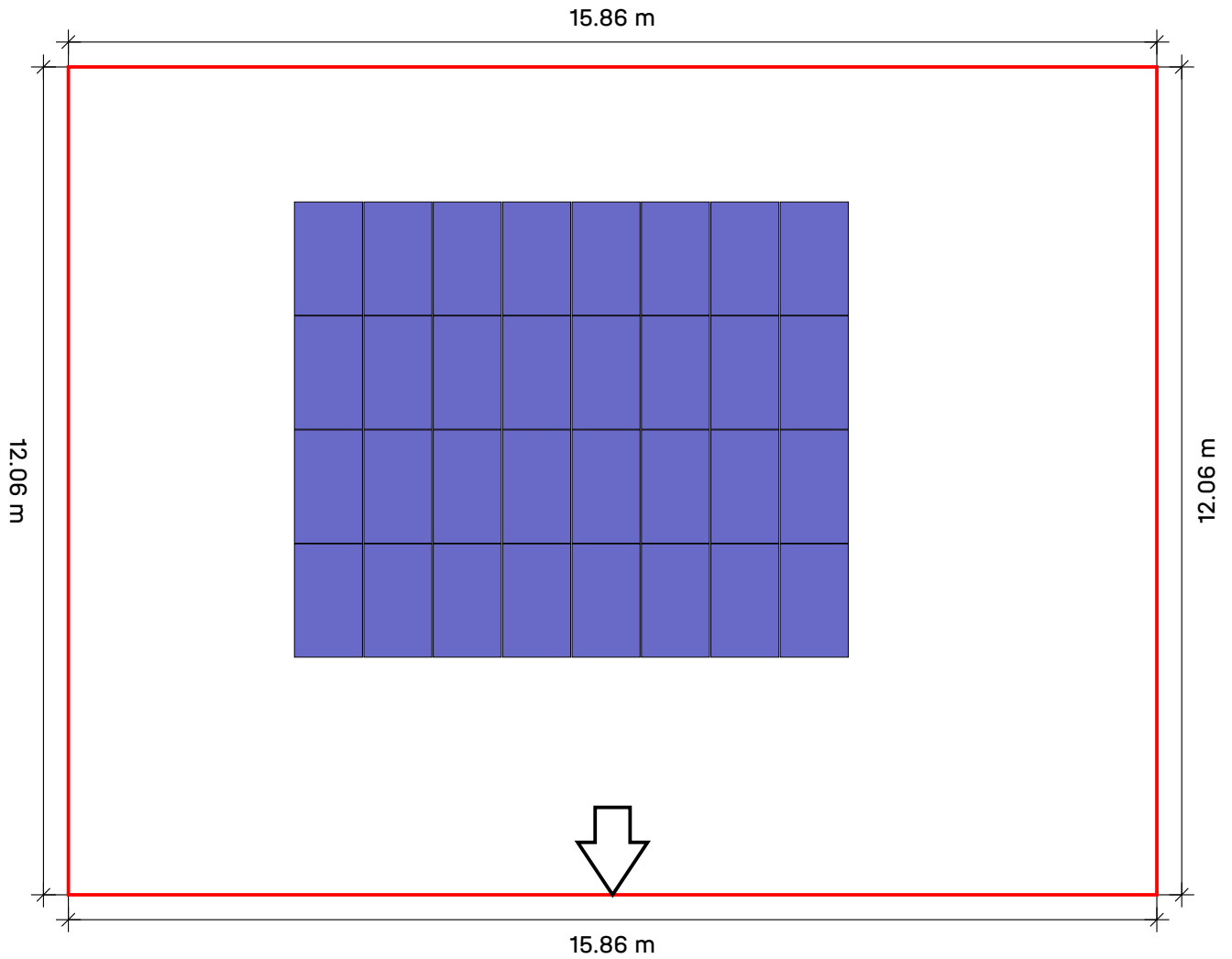
Position	Item no.	Item description	Quantity	Weight
1	2002226	Solid Clamp Rib-Roof Evolution	630	107.1 kg
2	1000637	T-Bolt 28/15 M10×20	630	11.5 kg
3	1000042	Hexagon flange nut M10	630	6.9 kg
4	1004767	SingleRail 36 End Cap	96	0.7 kg
5	2002870	K2 Solar Cable Manager	132	0.4 kg
6	2003229	SolidRail UltraLight; 4.40 m	59	181.8 kg
7	1004107	SolidRail UltraLight+Light RailConnector Set	52	11.7 kg
8	2003145	SingleRail Climber Set 36/50	312	20.9 kg
9	2004148	K2 Clamp MC 25-40 Black	216	14.1 kg
10	2004545	K2 Clamp EC 25-40 Black	96	6.9 kg
11	2001865	SingleRail 36; 2.10 m	132	212.8 kg
12	2003605	SingleRail 36/50 RailConnector Set	96	23.1 kg
Total				598.0 kg

Bereich 2





Bereich 2



Roof	System	Module	Height	Quantity	Total power
<u>Bereich 2</u>  Bitumen / Composite Shingles	<u>SingleRail</u>	AHT60M-250WBL 1,650×991×40 mm 250 Wp	15.00 m	32	8 kWp



Bereich 2 | Assembly plan

Base Rails

Type	Whole Rails		Rail cutting		
	Total Rail Length	Quantity 4.40 m	Part of Rail / Rest	Length	Rest
8*A	8.234 m	1*4.40 m	4.400	3.834 from 4.400	0.556

1 cm is viewed as lost for each cutting

Red numbers are leftover rails which will not be used any longer

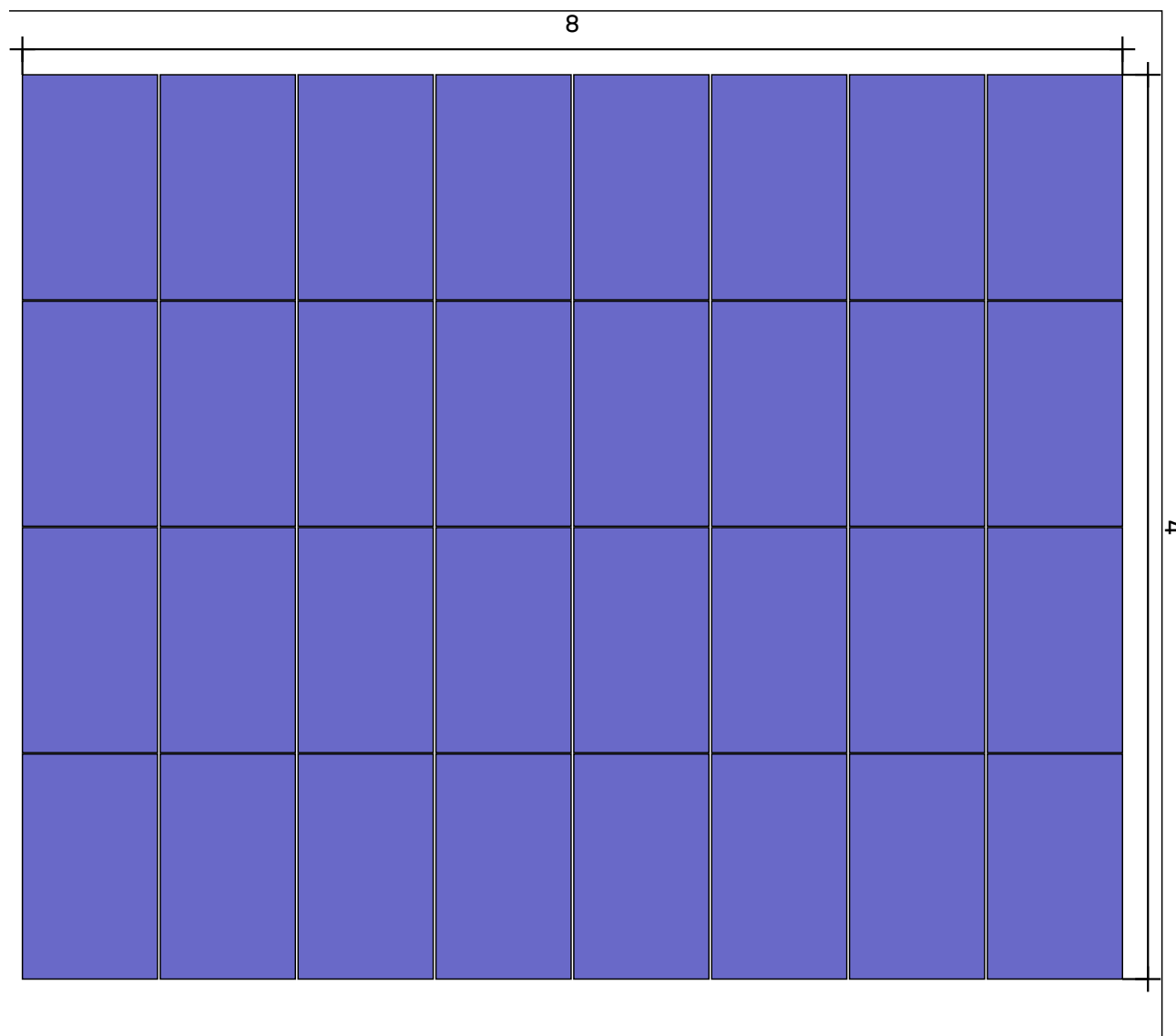
Fastener Spacing

Module	Array	Distance	maximum cantilever length [m]	maximum fastener spacing [m]
1	field area	1.00 m	0.482	1.576

Module arrays

Module array	Width[m]	Length[m]	Width in modules	Length in modules
1	8.07	6.63	8	4

Bereich 2 | Module array 1



Roof ② Module array ①

Mounting System

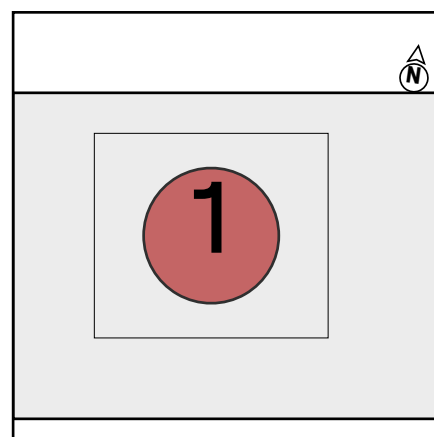
SingleRail

Module

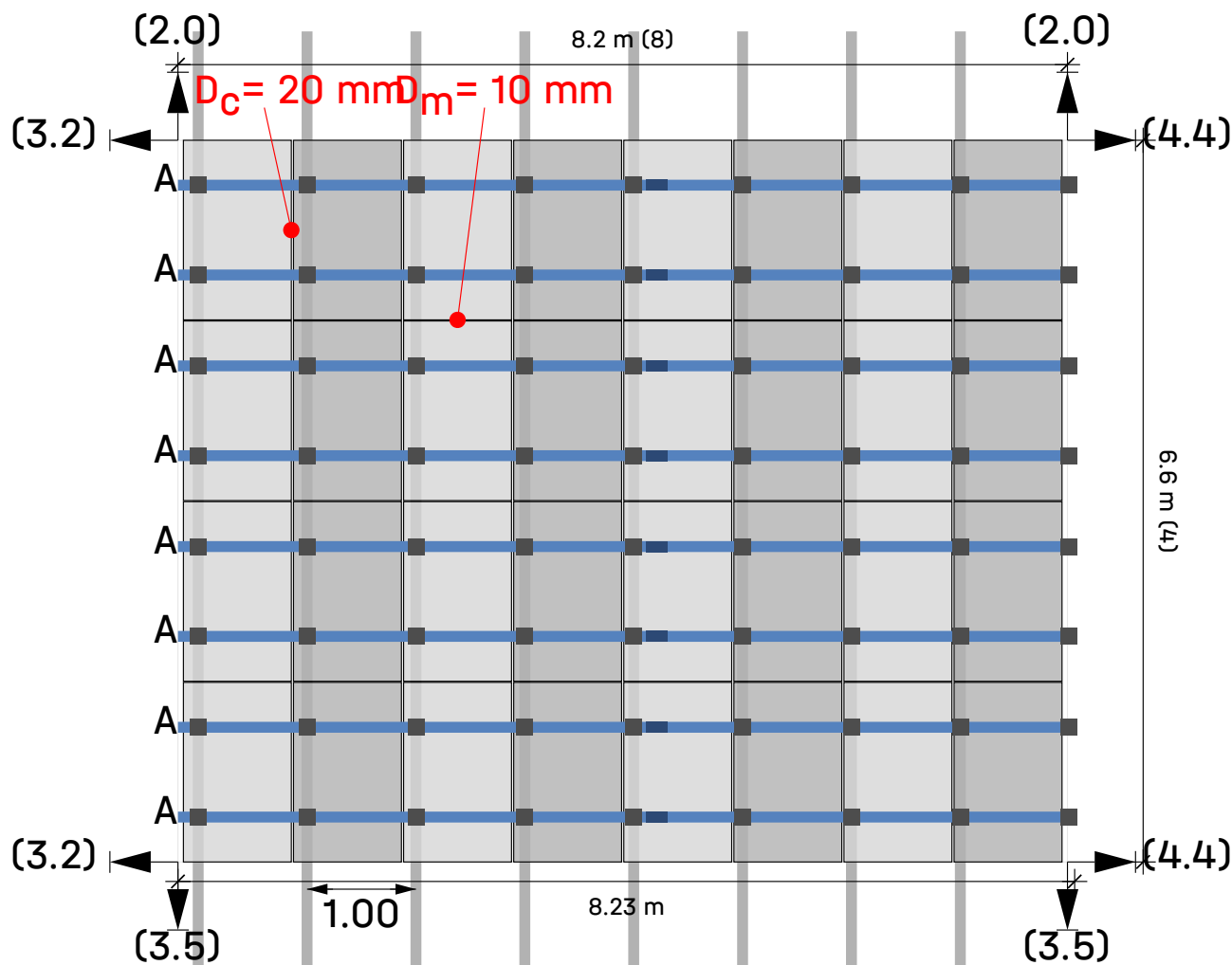
32(8 kWp) x
AHT60M-250WBL

Row spacing

1.66 m



Bereich 2 | Module array 1 | Module block 1

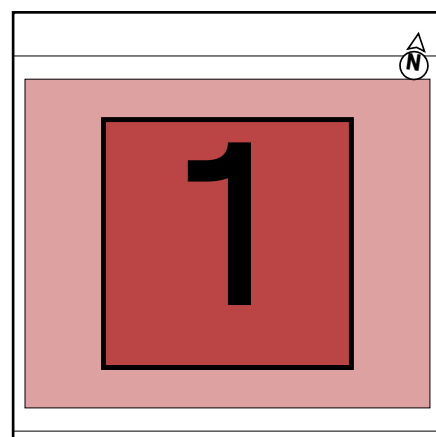


Roof ② Module array ① Module block 1

Modules $8 \times 4 = 32$

Legend

- Fastener
- Mounting rail: K2 SingleRail 36
- Distance to Roof Edge [m]
- D_c Distance for clamping between modules
- D_m Distance between modules





Results | Bereich 2

Roof	System	Module	Height	Quantity	Total power
<u>Bereich 2</u>  Bitumen / Composite Shingles	<u>SingleRail</u>	AHT60M-250WBL 1,650×991×40 mm 250 Wp	15.00 m	32	8 kWp

Module

Name	AHT60M-250WBL
Manufacturer	Ahead Renewable Energy Ltd.
Output power	250 Wp
Dimensions	1,650×991×40 mm
Weight	20.0 kg

Components

Fastener	SingleRail Splice Foot
Base rails	K2 SingleRail 36

Loads on modules (module dimensioning)

No. Module Array	Array	A-TrA [m ²]	ultimate state [Pa]				Serviceability [Pa]			
			Pressure ⊥	Pressure 	Uplift ⊥	Uplift 	Pressure ⊥	Pressure 	Uplift ⊥	Uplift
1	field area	1.64	1,799.7	144.5	-1,211.5	11.8	1,213.3	97.5	-762.7	11.8

Utilisation result

No. Module Array	roof areas	ultimate limit state			Usab.	Distances		maximum values	
		Pr σ[%]	CL σ[%]	Fst F[%]	Pr f[%]	Fst [m]	BR [m]	CL L _{max} [m]	Fst Fst D _{max} [m]
1	field area	40.3	7.0	28.1	24.2	1.000	---	0.482	1.576

Pr	Profile	Fst D _{max}	maximum fastener spacing [m]
Fst	Fastener	BR	Base Rail
σ	Stress	Usab.	serviceability limit state
f	Deflection	CL	Cantilever
F	Force		
CL/L _{max}	maximum cantilever length [m]		



Results | Bereich 2

Important Information

- The resistance values for the screws are based on a wood material quality of C24 ($\rho \geq 350 \text{ kg/m}^3$) and a $k_{\text{mod}} = 0.9$
- The structure was statically verified in accordance with Eurocode 9: Design of aluminium structures (DIN EN 1999-1-1:2021) and offers sufficient load-bearing capacity and stability for the loads specified in the chapter 'Maximum actions on the components'.
- Adjustment factor for wind load regarding service life period, f_W , is according to DIN EN 1991-1-4/NA, NDP for 4.2 (2P) note 5, table 3
- Adjustment factor for snow load regarding service life period, f_S , is according to DIN EN 1991-1-3/annex D, table 4
- The design rules comply with the Eurocode EN 1990 - Basis of structural design. (DIN EN 1990: 2010)
- The determination of snow loads is based on the National Annex to DIN EN 1991-1-3/NA - Snow Loads.
- The determination of wind loads is based on the National Annex to DIN EN 1991-1-4/NA - Wind Loads.
- Service life is recognised according to 'Eurocode EN 1991 - Action on structures, Snow loads' and 'Eurocode EN 1991 - Actions on structures, Wind actions'. Subject to the Building Regulations and for security-relevant reasons the installation has to be dismantled at the end of its service life.
- Failure consequence class is considered according to 'Eurocode EN 1990 - Basis of structural design'.
- The person responsible for the execution of the work must check the load assumptions made with the conditions on site. If deviations are found, the person who prepared the static calculation must be consulted immediately.
- Please note our General Terms and Conditions of Use (TCU-E) in their currently valid version, available at: <https://k2-systems.com/en/digital-services/general-terms-and-conditions-of-use-for-entrepreneurs-tcu-e/>
Please note in particular § 1, Special provisions for K2 Base, clause 3 ("Technical and specialist requirements on the Customer's premises."), § 6 ("Limitation of warranty") and § 7 ("Limitation of liability").
- All instructions and requirements of the static calculations must be observed.
- Compliance with the regulations and requirements of the building permit and the building trade associations must be ensured during the execution of all work.
- All connections and anchoring measures must be carried out using approved and standardized connecting elements or parts only.
- Subordinate components that have not been verified must be implemented structurally.
- The "General Rules" for corrosion protection design of the structure are strongly recommended for application during implementation.
- The structure may only be climbed by trained and authorized personnel wearing appropriate safety equipment. Unauthorized use of the structure poses a risk of falling.

Structural analysis report | Bereich 2

General information

Name	BDP-4863
Mounting System	SingleRail
Author	shobana Test account

Location information

Address	71 Renningen
Ground elevation	405.51 m

Roof information

Building height	15.00 m
Roof type	Gable roof
Roof pitch	5°
Fastening method	Fixed in rafter
Roof covering	Bitumen / Composite Shingles
Min. roof edge distance	0.00 m
Rafter Spacing	1.000 m
Rafter width	100.0 mm
set edge rafter left	No
Rafter spacing left	430.0 mm
set edge rafter right	No
Rafter spacing right	430.0 mm
counter lath thickness	40.0 mm
Rafter height	40.0 mm
deck material	Solid Wood
deck thickness	40.0 mm
Covering height	5.0 mm

Loads

Design method	DIN EN
Failure consequence class (CC)	CC2
Design working life	50 years
Terrain category	I/II - mixed profile rural area

Wind load

Wind load zone	1
Velocity pressure, 50 years	$q_{p,50} = 0.821 \text{ kN/m}^2$
Adjustment factor for service life	$f_w = 1.000$
Velocity pressure, 25 years	$q_{p,25} = 0.821 \text{ kN/m}^2$



Structural analysis report | Bereich 2

Roof areas

No. Module Array	Array	load impact area [m ²]	maxCpe	minCpe	wind pressure [kN/m ²]	wind suction [kN/m ²]
1	field area	10.00	0.200	-0.700	0.164	-0.575

Snow load

Snow load zone	2
Snow guard	No
Snow load on ground level	$s_k = 1.234 \text{ kN/m}^2$
Shape Coefficient for Snow	$\mu_i = 0.800$
Factor for roof pitch	$d_i = 0.996$
Snow load on roof, 50 year	$s_{i,50} = 0.983 \text{ kN/m}^2$
Adjustment factor for service life	$f_s = 1.000$
Snow load on roof, 25 years	$s_{i,25} = 0.983 \text{ kN/m}^2$

Dead Load

Weight of module	$G_M = 20.0 \text{ kg}$
Weight of mounting system per module	$= 2.60 \text{ kg}$
Module area	$A_M = 1.64 \text{ m}^2$
Dead weight of module per m ²	$= 12.23 \text{ kg/m}^2$
Dead weight of mounting system per m ²	$= 1.59 \text{ kg/m}^2$
Total Dead Load (excl. ballast) per m ²	$= 0.14 \text{ kN/m}^2$

Load Combinations

Ultimate limit state

Partial safety factor unfavourable permanent load	$\gamma_{G,sup} = 1.35$
Partial safety factor favourable permanent load	$\gamma_{G,inf} = 1.00$
Partial safety factor destabilising permanent load	$\gamma_{G,dst} = 1.10$
Partial safety factor stabilising permanent load	$\gamma_{G,stab} = 0.90$
Partial safety factor variable loads	$\gamma_Q = 1.50$
Combination coefficient with regards to wind	$\psi_{0,W} = 0.60$
Combination coefficient with regards to Snow	$\psi_{0,S} = 0.50$
Importance factor permanent	$\kappa_{FI,G} = 1.00$
Importance factor variable	$\kappa_{FI,Q} = 1.00$

Load case combination 01	$LCC\ 01_{uls} = \gamma_{G,sup} * \kappa_{FI,G} * G_k + \gamma_Q * \kappa_{FI,Q} * S_{i,n}$
Load case combination 02	$LCC\ 02_{uls} = \gamma_{G,sup} * \kappa_{FI,G} * G_k + \gamma_Q * \kappa_{FI,Q} * W_{k,Pressure}$
Load case combination 03	$LCC\ 03_{uls} = \gamma_{G,sup} * \kappa_{FI,G} * G_k + \gamma_Q * \kappa_{FI,Q} * (W_{k,Pressure} + \psi_{0,S} * S_{i,n})$



Structural analysis report | Bereich 2

Load case combination 04

$$\text{LCC 04_uls} = \gamma_{G,\text{sup}} * K_{\text{Fl,G}} * G_k + \gamma_Q * K_{\text{Fl,Q}} * (S_{i,n} + \psi_{0,W} * W_{k,\text{Pressure}})$$

Load case combination 06

$$\text{LCC 06_uls} = \gamma_{G,\text{inf}} * G_k + \gamma_Q * K_{\text{Fl,Q}} * W_{k,\text{Suction}}$$

Serviceability limit state

Combination coefficient with regards to wind

$$\psi_{0,w} = 0.60$$

Combination coefficient with regards to Snow

$$\psi_{0,s} = 0.50$$

Load case combination 01

$$\text{LCC 01_sls} = G_k + S_{i,n}$$

Load case combination 02

$$\text{LCC 02_sls} = G_k + W_{k,\text{Pressure}}$$

Load case combination 03

$$\text{LCC 03_sls} = G_k + W_{k,\text{Pressure}} + \psi_{0,s} * S_{i,n}$$

Load case combination 04

$$\text{LCC 04_sls} = G_k + S_{i,n} + \psi_{0,W} * W_{k,\text{Pressure}}$$

Load case combination 06

$$\text{LCC 06_sls} = G_k + W_{k,\text{Suction}}$$

Maximum load on modules (Mounting system dimensioning)

No. Module Array	Array	A-TrA [m ²]	ultimate state [kN/m ²]				Serviceability [kN/m ²]			
			Pressure ⊥	Pressure 	Uplift ⊥	Uplift 	Pressure ⊥	Pressure 	Uplift ⊥	Uplift
1	field area	10.00	1.800	0.145	-0.727	0.012	1.213	0.098	-0.440	0.012

Max. load on fastener

No. Module Array	Array	A-TrA [m ²]	ultimate state [kN]				Serviceability [kN]			
			Pressure ⊥	Pressure 	Uplift ⊥	Uplift 	Pressure ⊥	Pressure 	Uplift ⊥	Uplift
1	field area	10.00	1.633	0.131	-0.660	0.011	1.101	0.089	-0.399	0.011

Resistance Values of Components

Base Rails

Base Rails	A [cm ²]	I _y [cm ⁴]	I _z [cm ⁴]	W _y [cm ³]	W _z [cm ³]
K2 SingleRail 36	2.850	4.02	6.37	2.14	3.09

Fastener

Fastener	R _{D,Uplift,Perpendicular} [kN]	R _{D,Pressure,Perpendicular} [kN]	R _{D,Pressure,Parallel} [kN]
SingleRail Splice Foot	5.65	5.87	2.56
Selbstbohrende Blechschraube 6,8×140	2.66	-	1.17



Structural analysis report | Bereich 2

Utilisation result

No.	roof areas	ultimate limit state			Usab.	Distances		maximum values	
		Pr	CL	Fst	Pr	Fst	BR	CL	Fst
Module Array		σ [%]	σ [%]	F[%]	f[%]	[m]	[m]	L_{max} [m]	Fst D_{max} [m]
1	field area	40.3	7.0	28.1	24.2	1.000	---	0.482	1.576

Pr	Profile	Fst D_{max}	maximum fastener spacing [m]
Fst	Fastener	BR	Base Rail
σ	Stress	Usab.	serviceability limit state
f	Deflection	CL	Cantilever
F	Force		
CL/L_{max}	maximum cantilever length [m]		



Bereich 2 | Bill of material

Position	Item no.	Item description	Quantity	Weight
1	2004545	K2 Clamp EC 25-40 Black	16	1.1 kg
2	2003393	SingleRail Splice Foot	72	8.7 kg
3	2003527	Self Drilling Screw 6,8×140	72	1.9 kg
4	1002387	T-Bolt 20/12	72	0.7 kg
5	1000043	Hexagon flange nut M8	72	0.5 kg
6	2004148	K2 Clamp MC 25-40 Black	56	3.7 kg
7	1004767	SingleRail 36 End Cap	16	0.1 kg
8	2003523	BlackCover SingleRail 36	16	0.4 kg
9	2002870	K2 Solar Cable Manager	32	0.1 kg
10	2003222	SingleRail 36; 4.40 m	16	54.2 kg
11	2003605	SingleRail 36/50 RailConnector Set	8	1.9 kg
Total				73.4 kg



Bill of material

Position	Item no.	Item description	Quantity	Weight
1	2002226	Solid Clamp Rib-Roof Evolution	630	107.1 kg
2	1000637	T-Bolt 28/15 M10×20	630	11.5 kg
3	1000042	Hexagon flange nut M10	630	6.9 kg
4	1004767	SingleRail 36 End Cap	112	0.8 kg
5	2002870	K2 Solar Cable Manager	164	0.5 kg
6	2003229	SolidRail UltraLight; 4.40 m	59	181.8 kg
7	1004107	SolidRail UltraLight+Light RailConnector Set	52	11.7 kg
8	2003145	SingleRail Climber Set 36/50	312	20.9 kg
9	2004148	K2 Clamp MC 25-40 Black	272	17.8 kg
10	2004545	K2 Clamp EC 25-40 Black	112	8.0 kg
11	2001865	SingleRail 36; 2.10 m	132	212.8 kg
12	2003605	SingleRail 36/50 RailConnector Set	104	25.1 kg
13	2003393	SingleRail Splice Foot	72	8.7 kg
14	2003527	Self Drilling Screw 6,8×140	72	1.9 kg
15	1002387	T-Bolt 20/12	72	0.7 kg
16	1000043	Hexagon flange nut M8	72	0.5 kg
17	2003523	BlackCover SingleRail 36	16	0.4 kg
18	2003222	SingleRail 36; 4.40 m	16	54.2 kg
Total				671.4 kg



Thank you for choosing a K2 mounting system.

Systems from K2 Systems are quick and easy to install.

We hope these instructions have helped.

Please contact us with any questions or suggestions for improvement.

Our contact data:

k2-systems.com/en/contact

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